

JAMES FLORENCE CONALES

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Quezon City, Metro Manila, Philippines

SOFTWARE ENGINEER

Fintech-specialized Backend Engineer with a focus on high-stakes payment infrastructure and secure cloud architecture. Proven track record of designing greenfield REST platforms and orchestrating complex EMI migrations while maintaining strict backward compatibility. Expert in GCP-native deployments and security remediation, having standardized organizational encryption protocols using Google Cloud KMS to resolve high-risk audit findings. Dedicated to building resilient, observable systems that balance rapid business growth with technical debt management.

PROFESSIONAL EXPERIENCE

PETNET, Inc

Software Engineer

June 2024 - Present

Software Engineer specializing in high-availability fintech systems and secure cloud architecture. Proven track record of leading end-to-end payment migrations and architecting greenfield backend platforms on GCP. Expert in transforming legacy monolithic systems into scalable, security-compliant microservices.

- **System Ownership & Reliability:** Maintain end-to-end technical ownership of the PERAHUB Mobile App backend, serving as the Primary Technical Responder for all production incidents and customer-facing concerns; ensured 99.9% service availability through proactive monitoring and rapid hotfix deployment.
- **Full-Stack Architecture (DFP):** Design & Architectural Oversight (DFP): Conceptualized and architected a greenfield Digital Forex Platform using a Nuxt 3/Express full-stack framework; designed the state-machine logic for document approval workflows and served as the Technical Authorizer for all production code and FOREX API integrations.
- **Strategic Migrations & Leadership:** Acted as Technical Lead for a high-pressure, 5-month EMI migration (EON to NetBank) and the Western Union modernization, successfully transitioning legacy SOAP services to REST while ensuring 100% contract stability for third-party vendors.
- **Security & Compliance Standardizations:** Spearheaded organizational adoption of Google Cloud KMS for cryptographic key management, remediating high-risk VAPT vulnerabilities and engineering secure API layers with request signing and payload encryption.
- **Platform Modernization:** Drove the transition of legacy Laravel 4.2 monoliths to high-performance Node.js (v22) environments on GCP Cloud Run, optimizing system latency and developer productivity across 10+ business-critical systems.
- **Production Incident Leadership:** Author comprehensive technical "Runbooks" for troubleshooting partner-side API failures and production bottlenecks, significantly lowering the Mean-Time-To-Recovery (MTTR) across the mobile engineering department.
- **Engineering Process Standardization:** Established a standardized "Security-First" development lifecycle by creating reusable encryption modules and middleware templates, reducing the onboarding time for new payment integrations by 30%.
- **System Design & Architectural Documentation:** Spearheaded the creation of technical architecture documentation and Data Flow Diagrams (DFDs) for the Perahub Mobile ecosystem, providing a source of truth that bridged communication between non-technical stakeholders and Solution Architects.
- **Cross-Platform Feature Delivery:** Led the development of high-impact features including QRPH support and automated payment schedulers within the Node.js ecosystem, ensuring these new services maintained 100% backward compatibility with legacy mobile frontend versions.
- **Infrastructure & Observability:** Architected automated CI/CD lifecycles using GitLab CI and Docker, enforcing "Security as Code" while implementing centralized logging via GCP Log Explorer for enhanced audit traceability.

TECHNICAL SKILLS

Backend & Architecture

- Core: Node.js (v22), PHP (7.4–8.3), Express.js, Laravel, Lumen.
- Design Patterns: Microservices Architecture, REST & SOAP API Design, Event-Driven Design, State Machine Logic (Approval Workflows).
- System Design: Data Flow Diagrams (DFDs), Database Schema Design, API Contract Preservation, Idempotency, and Rate Limiting.

Cloud & Infrastructure (GCP Certified)

- Compute & Orchestration: Google Cloud Run (Serverless), Docker (Containerization).
- DevOps/CI/CD: GitLab CI/CD (Pipeline Automation), Cloud Build, Standardized Environment Management.
- Storage & Messaging: PostgreSQL, MySQL, Google Cloud Storage, Pub/Sub (Ongoing).

Security & Compliance

- Cryptography: Google Cloud KMS (Key Management), Request/Response Signing (HMAC/RSA), Payload Encryption at Rest/Transit.
- Identity: IAM (Principle of Least Privilege), RBAC (Role-Based Access Control), OAuth2, Secure Session Management.
- Remediation: VAPT Finding Resolution, Audit-Driven Security Hardening.

Full-Stack & Observability

- Frontend (Modern Stack): Nuxt.js (SSR), Vue.js, React, Next.js, Tailwind CSS.
- Observability: Centralized Logging (GCP Log Explorer), Middleware-based Tracing, Incident Triage (MTTR reduction).

EDUCATION

FEU Institute of Technology 2019 - 2024

Bachelor of Science in Computer Science Major in Software Engineering (BSCSSE)

Cum Laude